



DEPARTMENT OF COMPUTER APPLICATIONS

Academic Year: 2026-2027

MCA Semester 3

A250201/CA330V MINI PROJECT

PROJECT DIARY

WEATHER GUARD

Register Number:

Name:

Batch: MCA 2025-2027

Project guide: Mr. Sivadas T Nair

Problem Statement

- Existing weather applications provide basic data like temperature and forecasts but lack integrated, real-time disaster alerts and actionable safety guidance. During events such as floods or cyclones, users must rely on multiple sources (weather apps, news, government sites), leading to delays and confusion.
- Additionally, current systems do not offer personalized risk assessment based on location, nor do they provide essential emergency preparedness support like safety guidelines or response actions.
- Therefore, there is a need for a unified platform that delivers real-time weather updates, integrates reliable disaster alerts, provides location-based risk assessment, and offers clear emergency guidance.
- The proposed system, WeatherGuard, addresses these gaps by combining weather data, alerts, and safety support into a single user-friendly application to enhance awareness and response during critical situations.

Objective No.	Objective Description
1	To provide real-time weather updates including temperature, humidity, and forecasts.
2	To integrate disaster alerts from reliable sources such as IMD and NDMA.
3	To implement location-based risk assessment for better awareness of weather severity.
4	To deliver clear and actionable emergency guidance during disasters.
5	To reduce user dependency on multiple information sources by offering a unified platform.
6	To enhance user preparedness through safety guidelines and response measures.
7	To ensure a user-friendly interface for easy access to critical information.

PROJECT PLAN

Week	Date	Task
1	06-07-2026 to 10-07-2026	• Literature survey, problem identification • Research weather apps, IMD/NDMA systems, identify gaps finalize project idea • Synopsis preparation & submission
2	13-07-2026 to 17-07-2026	• Synopsis discussion & approval
3	20-07-2026 to 24-07-2026	• SRS, SDD, test plan preparation
4	27-07-2026 to 31-07-2026	• Prepare documents + basic UI design
5	03-08-2026 to 07-08-2026	• Setup backend (Spring Boot), database, create User module + basic APIs
6	10-08-2026 to 14-08-2026	• Implement JWT authentication + start weather API integration
7	17-08-2026 to 21-08-2026	• Complete weather APIs + start frontend (React setup, login UI)
8	31-08-2026 to 04-09-2026	• Show working: login + weather dashboard (basic), partial frontend
9	07-09-2026 to 11-09-2026	• Implement alerts module + start Smart Risk Indicator logic
10	14-09-2026 to 18-09-2026	• Complete Risk Indicator + location-based personalization
11	21-09-2026 to 25-09-2026	• Build Emergency Assistant + integrate all modules
12	28-09-2026 to 02-10-2026	• Testing, bug fixing, UI improvement, prepare draft report
13	05-10-2026 to 09-10-2026	• Finalize project, deploy (optional), submit report + code
14	12-10-2026 to 16-10-2026	• Prepare PPT, demo flow, viva answers, explain architecture

PROGRESS REPORT – WEEK 1

During this week, a literature survey was conducted to study existing weather forecasting and disaster alert systems. It was observed that most applications provide basic weather details but lack integration with official disaster alerts and emergency preparedness guidance. Users often rely on multiple sources such as weather apps, news, and government websites (IMD, NDMA), which can cause delays and confusion during emergencies.

Information was gathered from weather APIs like Open-Meteo, government sources such as IMD and NDMA, and existing applications. Suitable technologies identified include React (frontend), Spring Boot (backend), and MySQL (database). It was noted that weather data can be fetched via APIs, while alerts can be integrated or simulated if required.

Based on the study, it was suggested to develop a centralized platform combining weather updates, disaster alerts, and preparedness guidance. Key features include a Smart Risk Indicator, location-based services, and an Emergency Preparedness module.

In conclusion, the week focused on understanding the problem and defining a solution. The proposed system, WeatherGuard, aims to improve user awareness and safety during adverse weather conditions.

Weekly Assessment Score based on deliverables and work progress

Task Planning, Timely Completion and Adherence to Schedule (5 marks)	
Development Progress and Application of Tools, Techniques and Methodologies (5 marks)	
Incorporation of Guide's Suggestions (5 marks)	
Total Marks (15 marks)	

Next Meeting Date:

Signature of the Guide:

PROGRESS REPORT – WEEK 2

During this week, the focus was on defining the problem statement and objectives of the system. It was observed that early identification of requirements helps avoid confusion during development. The system was structured into key modules such as User Management, Weather Data, Alerts, and Risk Indicator. The importance of using reliable APIs for real-time data and selecting a scalable technology stack was also recognized.

Information gathered included functional and non-functional requirements, REST API concepts, and suitable technologies like Spring Boot, React, and MySQL. The role of proper requirement analysis and SRS documentation in system development was clearly understood.

Based on this, it was suggested to clearly define requirements, organize the system into manageable modules, use reliable APIs, choose efficient technologies, and maintain proper documentation.

In conclusion, this week focused on requirement analysis and planning. The problem statement and objectives were finalized, establishing a clear direction for further design and development.

Weekly Assessment Score based on deliverables and work progress

Task Planning, Timely Completion and Adherence to Schedule (5 marks)	
Development Progress and Application of Tools, Techniques and Methodologies (5 marks)	
Incorporation of Guide's Suggestions (5 marks)	
Total Marks (15 marks)	

Next Meeting Date:

Signature of the Guide:

PROGRESS REPORT – WEEK 3

During this week, the focus was on preparing the Software Requirements Specification (SRS) and Software Design Document (SDD). It was observed that proper documentation helps clearly define system functionality, architecture, and user interactions. The importance of a well-structured design and module-wise organization was also recognized.

Information gathered included SRS and SDD structure, system architecture (frontend, backend, API integration), module design, project planning, and the use of diagrams like ER and DFD. It was understood that standard documentation practices improve project clarity and quality.

Based on this, it was suggested to maintain structured documentation, use diagrams for better representation, divide the project into phases, ensure consistency between requirements and design, and regularly review documents.

In conclusion, this week focused on documentation and system design. SRS and SDD were prepared, and a project plan was created, providing a clear roadmap for development.

Weekly Assessment Score based on deliverables and work progress

Task Planning, Timely Completion and Adherence to Schedule (5 marks)	
Development Progress and Application of Tools, Techniques and Methodologies (5 marks)	
Incorporation of Guide's Suggestions (5 marks)	
Total Marks (15 marks)	

Next Meeting Date:

Signature of the Guide: